

Computer Network LAB-01 REPORT.

Experiment Title:-

Study of LAN Topology, Network Devices, and Address configuration.

Objective:-

To study LAN topology, identify networking devices, analyze wired and wireless communication, measure network speed, and obtain system network.

Problem Statement

1) Draw the Network Topology:

→ Identify and draw the LAN topology of your organization (e.g:- Star, Bus, Mesh).

→ Clearly label all components such as Router, Switch, Access point, and end devices (PCs, laptops, etc)

QoI: The college campus network examined during this study includes the following components:

→ Layer-3 core switches installed in the server room for network management and routing.

→ PoE (Power over Ethernet) switches used to supply power and connectivity to network devices.

→ Wireless Access Points (APs) for providing Wifi connectivity across the campus.

→ Ethernet ports available in classrooms and laboratories for wired network access.

The college campus network uses a Hybrid Hierarchical Topology that combines:

- Tree Topology.
- Extended Star Topology
- Ring Topology (in server room)

The network is divided into three layers:

1. Core layer.
2. Distribution layer.
3. Access layer.

The Core layer contains networking devices such as firewall, router, layer-3 switches, patch rack, and servers.

The Distribution layer consists of four switches placed in different rooms/floors of the main building for network distribution. (R7 - (212), R1 - (301), R6 - (112), (R2) - (201))

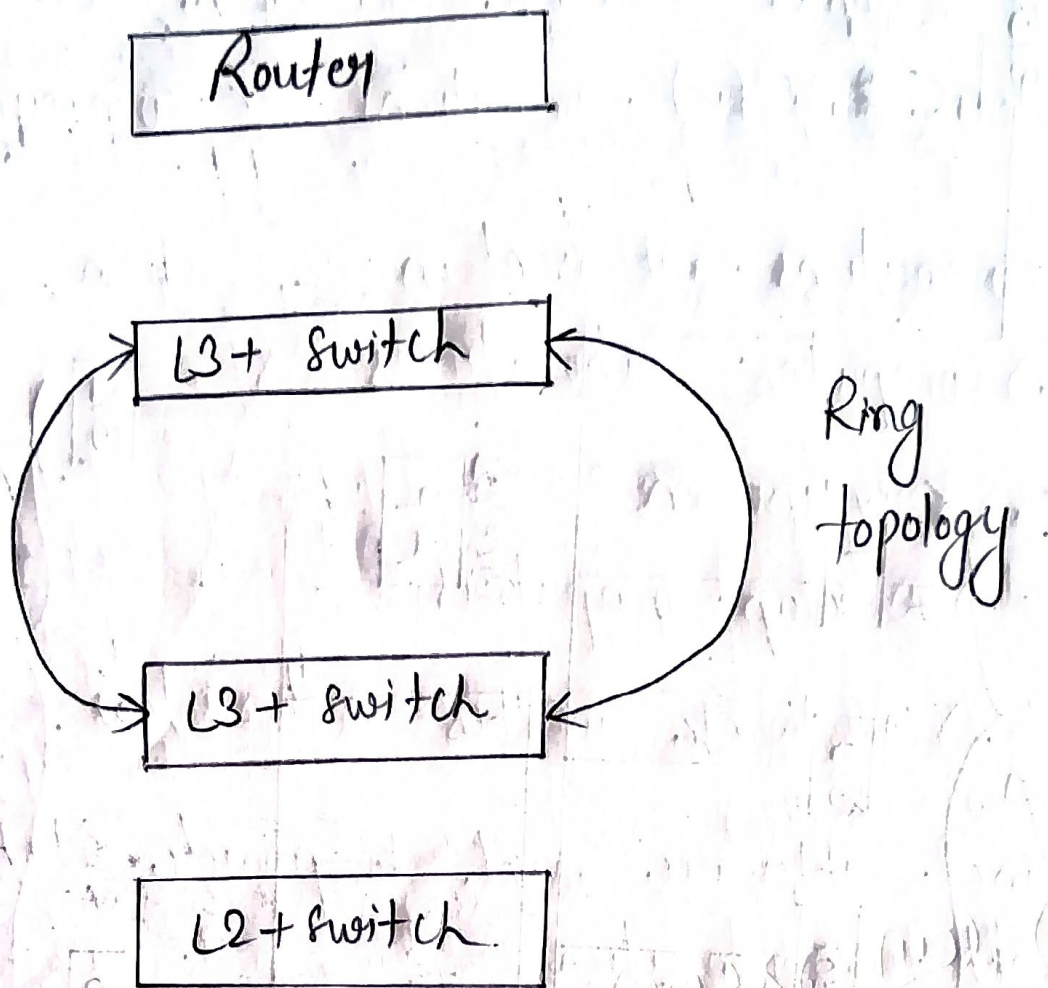
The Access layer contains three switches located in different rooms of the main building:

- R1 - Room No. 301 (3rd floor)
- R4 - Room No. 05 (Ground floor)
- R5 - Room No. 18 (Ground floor)

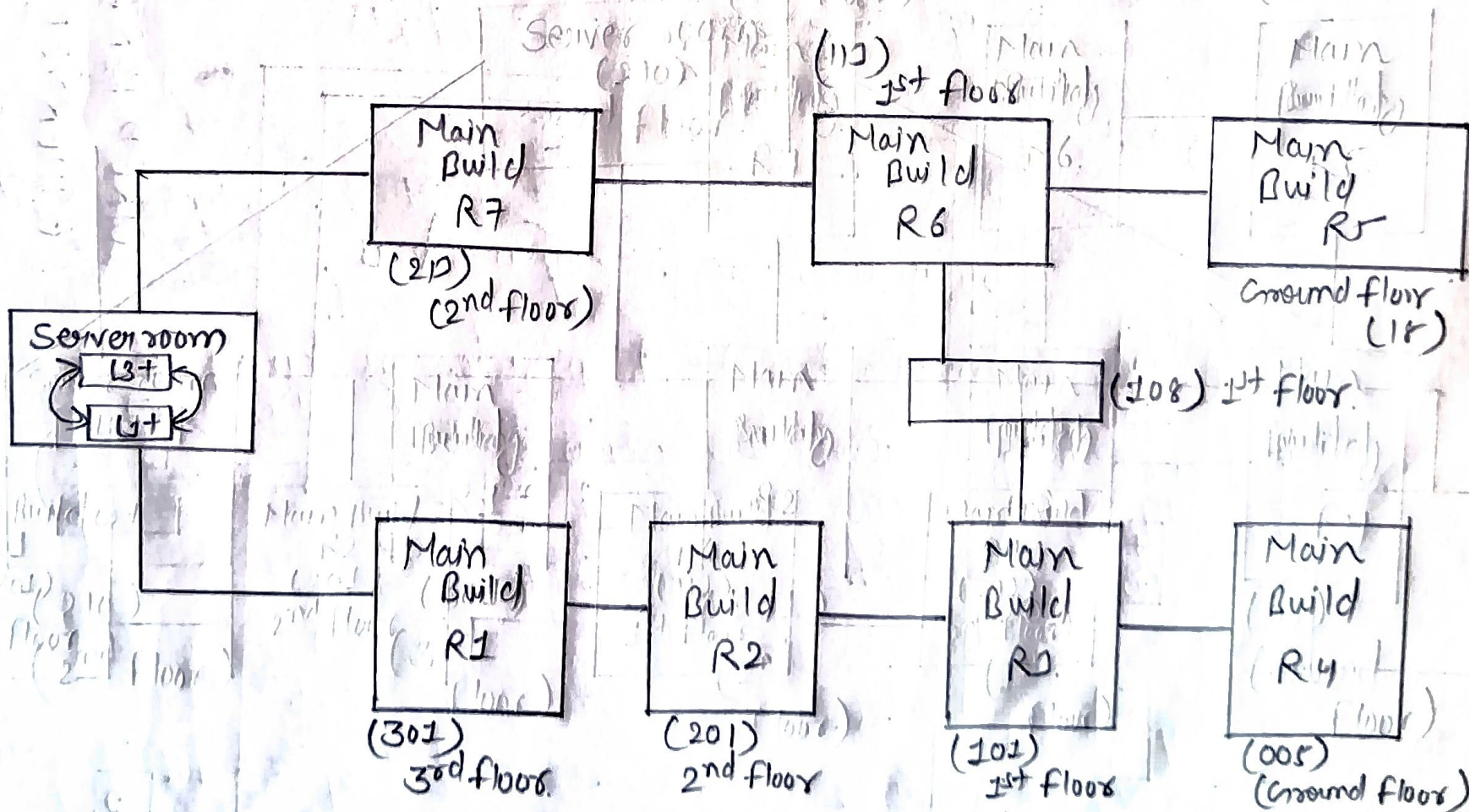
Access layer directly connect:

- ① Ethernet Port
- ② Wireless AP
- ③ Lab System
- ④ Outdoor wireless AP (ex - ois)

Server Room Architecture / Topology :-



Topology

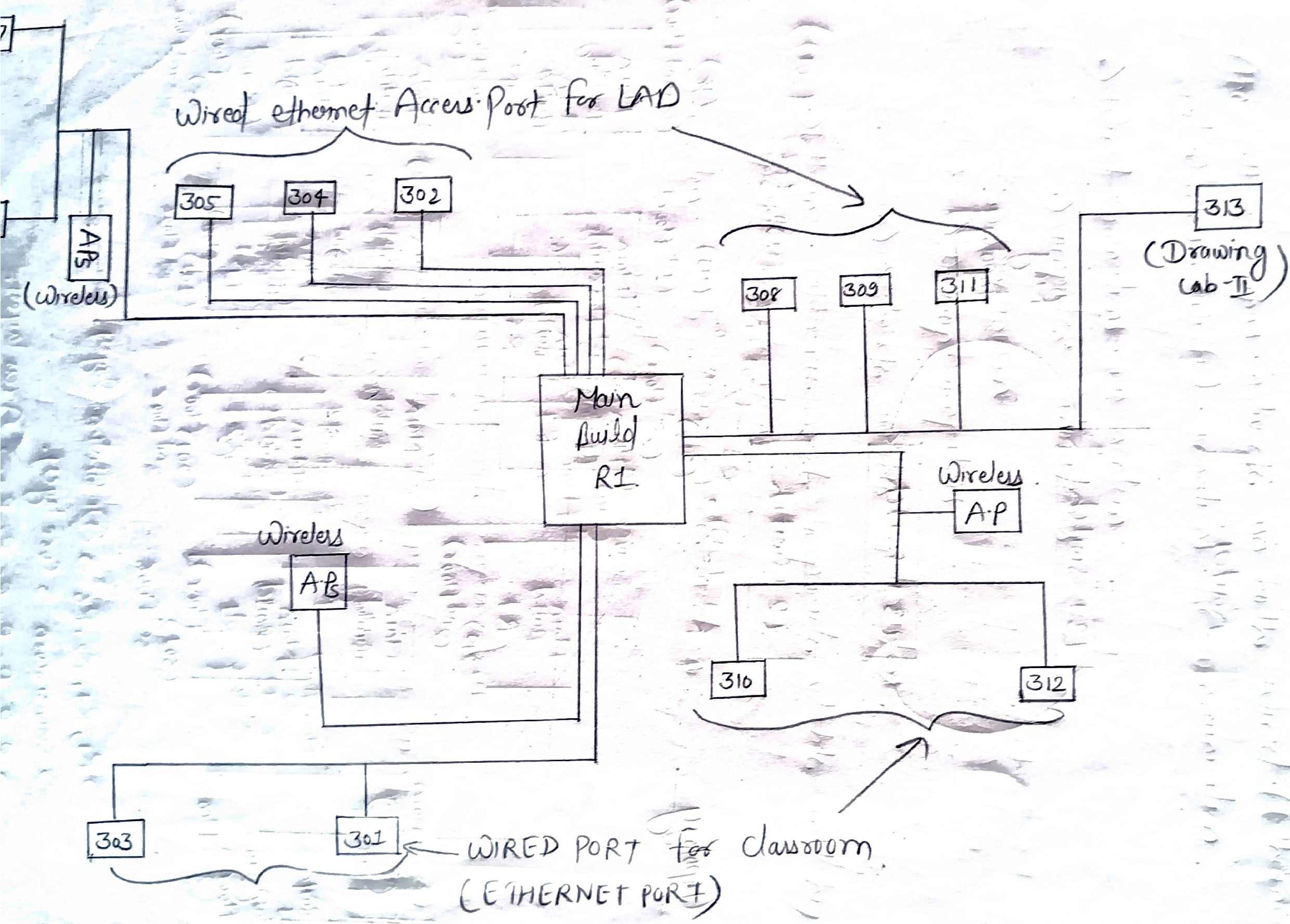


→ R₁, R₇, R₂, R₆ → Distribution layer.

→ Server room → Core layer.

→ R₁, R₂, R₃, R₄, R₅, R₆, R₇ → Access layer.

Access Layer: Switch (300 + 1000 + 301)



- ② Identify Network Devices:-
- List all networking devices used in the LAN
 - Mention the function of each device.
- ⇒ The following networking devices are used in the college LAN:

Device	function
Router	Connects the local network to the internet and manages data routing.
Layer-3 switch	Perform switching and routing operations within the network.
PoE switch	Provides both power and network connectivity to devices like Access points.
Wireless Access Point (AP)	Provides Wi-Fi connectivity to wireless users.
Firewall	Protects the network from unauthorized access and cyber threats.
Server	Stores and manages network resources and services.
Patch Panel Ethernet Cable	Used for wired communication between devices.
End Devices (PCs/Laptops)	Used by users to access network resources and Internet services.

③ Analyse Connection types:-

The college campus network uses both Wired (Ethernet) and Wireless (Wi-fi) connections for communication.

.) Wired Connections are used in:-

.) Computer laboratories

.) Server room

.) Desktop computers

.) Network switches and routers.

.) Ethernet cables are used to provide stable and high speed communications.

.) Wireless Connection (Wi-fi) are used in:-

.) Laptops

.) Mobile phones

.) Smart phones

.) Classrooms and campus Areas.

.) Wi-fi Access Points (APs) provide wireless internet connectivity across the campus.

\Users\HP>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : *Media disconnected*
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : *Media disconnected*
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : *fe80::28e1:6fa4:a826:8e58%10*
IPv4 Address. : *10.85.21.219*
Subnet Mask : *255.255.248.0*
Default Gateway : *10.85.16.1*

Ethernet adapter Ethernet:

Media State : *Media disconnected*
Connection-specific DNS Suffix . :

Windows IP Configuration

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::3b70:566e:8ccc:9a24%61
IPv4 Address. : 10.17.245.119
Subnet Mask : 255.255.255.0
Default Gateway : 10.17.245.68

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

find the link speed of:

- Wired network (if available)
- Wireless network.

• Mention the tools/commands used.

Ans:- Link speed is shown in the images in the next page of both wired and wireless.

The tools/commands used to check the speed using:-

1) Task Manager

2) Command Prompt.

Command Used

cmd

netsh wlan show interfaces.

ere is 1 interface on the system:

Name	: Wi-Fi
Description	: Intel(R) Wi-Fi 6E AX211 160MHz
GUID	: 806a7e9f-b1d4-48f4-a4d2-d925025c7e5c
Physical address	: 6c:2f:80:09:ec:73
Interface type	: Primary
State	: connected
SSID	: Galaxy A14 5G C6BE
AP BSSID	: b0:19:21:30:b5:bd
Band	: 5 GHz
Channel	: 40
Connected Akm-cipher	: [akm = 00-0f-ac:08, cipher = 00-0f-ac:04]
Network type	: Infrastructure
Radio type	: 802.11ax
Authentication	: WPA3-Personal
Cipher	: CCMP
Connection mode	: Auto Connect
Receive rate (Mbps)	: 360
Transmit rate (Mbps)	: 216
Signal	: 65%
Rssi	: -72
Profile	: Galaxy A14 5G C6BE
QoS MSCS Configured	: 0
QoS Map Configured	: 0
QoS Map Allowed by Policy	: 0

nce

SAMSUNG Mobile USB Remote NDIS Network Dev

CPU
9% 1.55 GHz

Memory
6.7/15.8 GB (42%)

Disk 0 (C: D:)
SSD (NVMe)
%

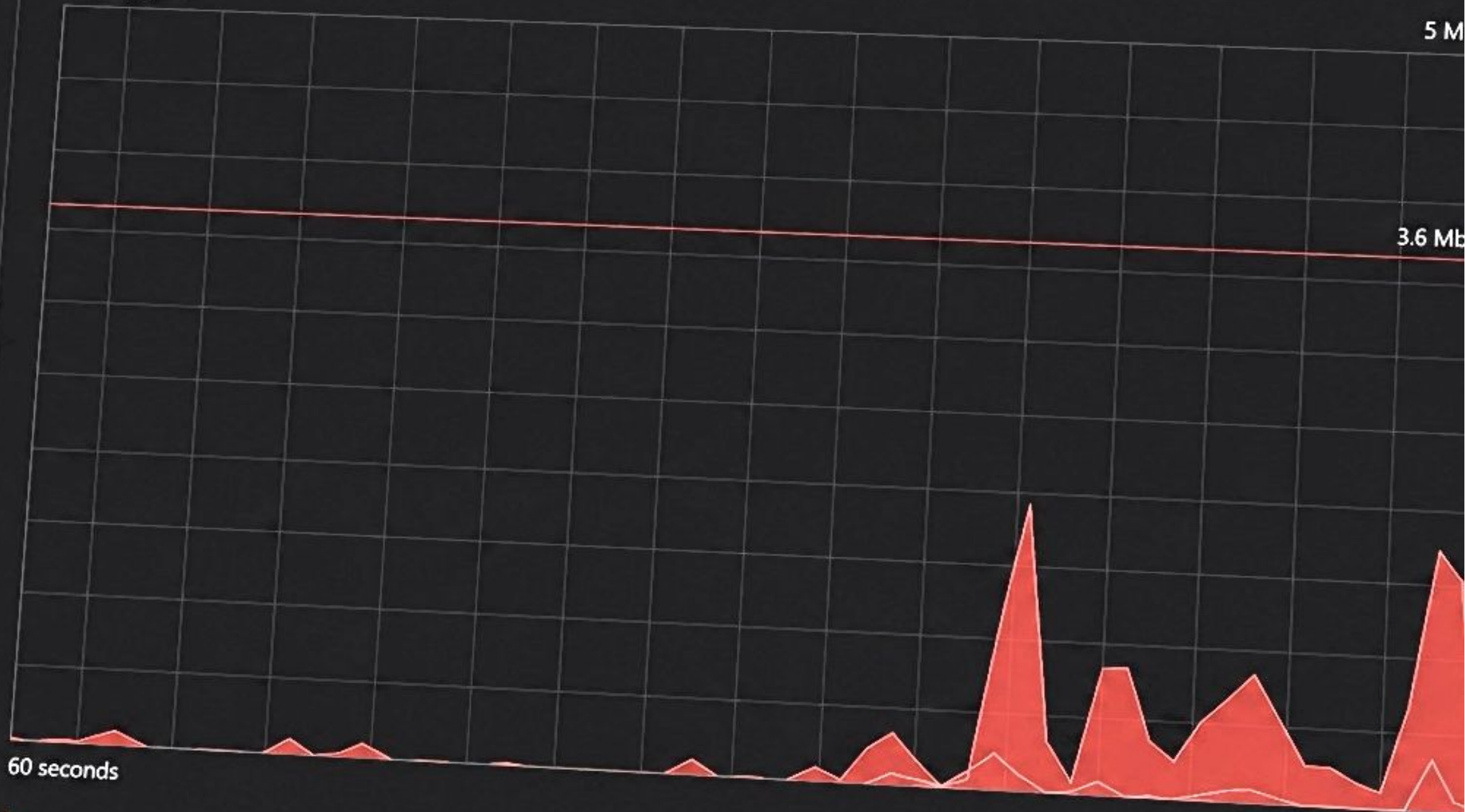
Ethernet
24.0 R: 272 Kbps

GPU 0
NVIDIA GeForce R...
(R) UHD Grap...

GPU 1
NVIDIA GeForce R...
(R) UHD Grap...
41 °C)

Ethernet

Throughput



Send

24.0 Kbps

Receive

272 Kbps

Adapter name: Ethernet 3

Connection type: Ethernet

IPv4 address: 10.17.215.119

IPv6 address: fe80::4e21:df8d:27c2:7686%61

CPU
% 1.58 GHz

Memory
/15.8 GB (39%)

Storage
C: (C: D:)
(NVMe)

Network
R: 392 Mbps

Graphics
UHD Grap...

GPU
GeForce R...

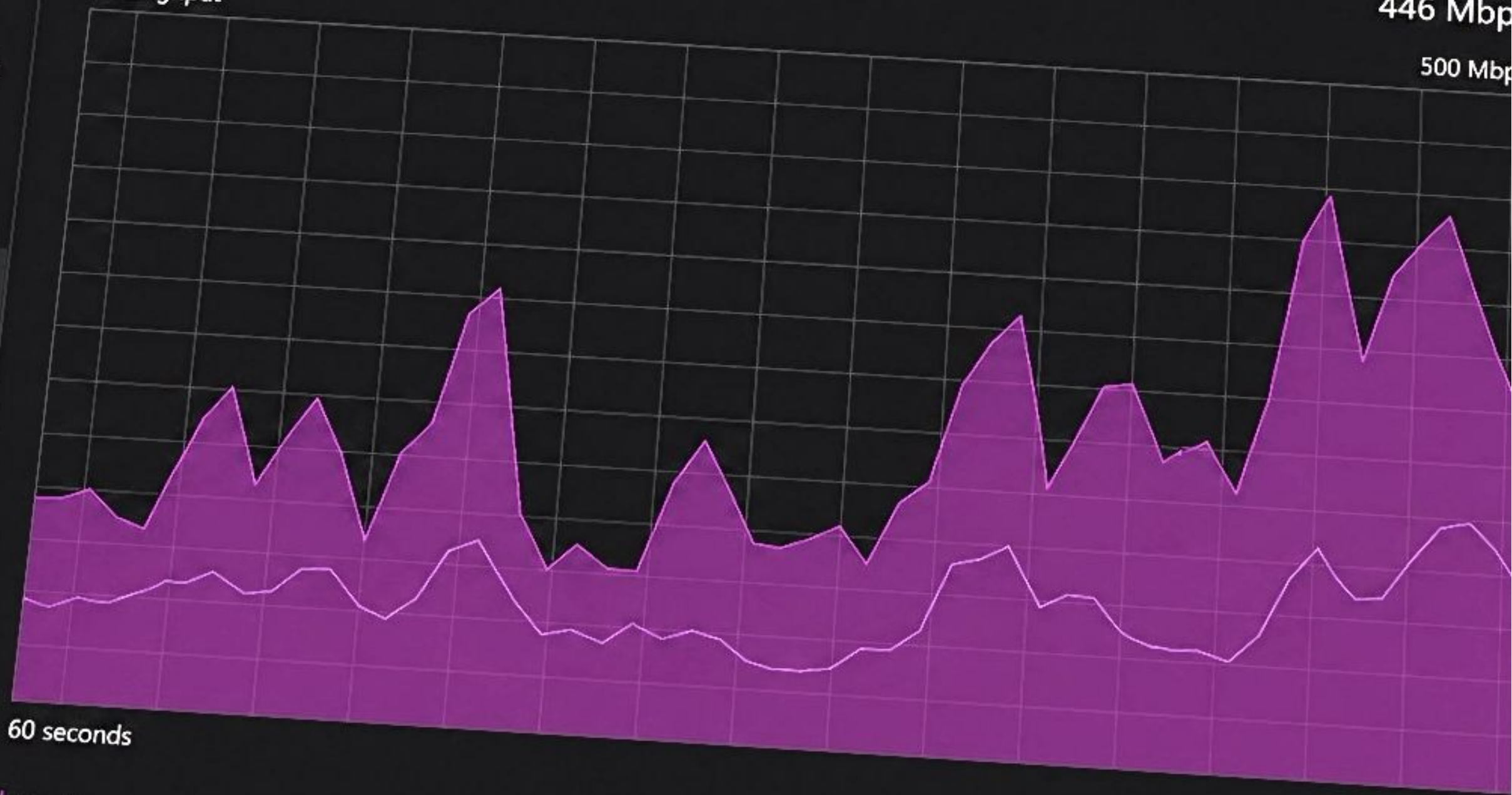
MediaTek Wi-Fi 5 MT7921 Wireless LAN Card

Wi-Fi

Throughput

446 Mbps

500 Mbps



60 seconds

Send

54.3 Mbps

Receive

392 Mbps

Adapter name:

Wi-Fi

SSID:

Galaxy A14 5G C6BE

Connection type:

802.11ac

IPv4 address:

10.85.21.219

IPv6 address:

fe80::28e1:6f4l:a826:8e58%10

Signal strength:



Link speed (B...

- Q find system · Network Details :-
- Determine and record the following of your Computer
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
 - Mention the command(s) used.

Ans: The record like MAC Address, IP Address (IPv4) and Subnet Mask shown in next page.

The command(s) used :-

→ to find IP Address and Subnet Mask
→ `ipconfig`

→ To find MAC Address.
→ `getmac`

```
\Users\HP>getmac
```

```
Physical Address
```

```
Transport Name
```

```
00-0C-2F-80-09-EC-73
```

```
\Device\Tcpip_{806A7E9F-B1D4-48F4-A4D2-D925025C7E5C}
```

```
08-00-28-C5-C8-5B-2D-E9
```

```
Media disconnected
```

⑥ → Difference between wired and wireless speed :-
⇒ There are various difference between wired and wireless speed :-

Wired Network	Wireless Network
→ Uses Ethernet cable for communication.	→ Uses wi-fi/radio signals.
→ Generally provides higher speed.	→ Speed may fluctuate.
→ More stable and reliable.	→ Affected by signal interference.
→ Lower latency.	→ Higher latency.
→ Better for gaming and large file transfer.	→ Better for mobility and convenience.
→ Less affected by distance.	→ Speed decreases with distance.

→ Role of subnet mask in networking :-

A subnet mask is used to divide an IP address into two parts :-

- Network portion
- Host portion

It helps devices identify whether another device is on the same network or a different network.

functions of Subnet Mask :-

- Identifies the network and host address
- Controls network traffic
- Helps in proper routing of data.
- Improves network management and security.

Examples:-

IPv4 Address : 10.17.245.119

Subnet Mask : 255.255.255.0

In this example :-

→ 10.17.245 is the Network ID

→ 119 is the Host ID.

indicated by the subnet mask 255.255.255.0

Conclusion:-

The experiment helped in understanding LAN topology, networking devices, and network configuration parameters. Different connection types such as wired and wireless networks were studied. Commands were used to identify IP address, MAC address, subnet Mask, and network speed. The experiment provided practical knowledge of computer networking fundamentals.